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Edge-Based vs Pairwise Closure Models

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Introduction

Two leading frameworks for modelling SIR-type epidemics on networks while avoiding stochastic simulation are:

1. Edge-Based Compartmental Models (EBCMs) — Miller's PGF-based ODEs that track the probability that a randomly chosen *test edge* has not yet transmitted infection (Miller 2011).
2. Pairwise (moment-closure) models — node and edge population fractions coupled to a third-order *triple* term that must be closed by an approximation (Keeling 1999).

EBCMs are exact on the configuration model in the large-`N` limit; pairwise models with a triple closure are approximations whose accuracy depends on the network structure and the closure choice.

In this vignette we use the disambiguating aliases to load both packages in one session and run two matched comparisons:

1. Regular network, $k = 5$. EBCM uses a degenerate PGF $\psi(z) = z^5$ (every node has degree 5); pairwise uses `regular_network(5)`.
2. Poisson network, $\kappa = 5$. EBCM uses `poisson_pgf(5.0)`; pairwise uses `erdos_renyi_network(5)` (a `HeterogeneousNetwork` whose degree distribution is Poisson(5)).

Within each scenario we recompute the per-edge transmission rate so the network epidemic has the same $R_0 = 2$ anchor as the well-mixed reference.

Setup

Both packages export `sir_model`, `default_initial_conditions`, `compartment`, `population_fraction`, etc. To avoid namespace collisions we import `NodeBasedModels` under an alias and use Edge's API by default.

```
using EdgeBasedModels
import NodeBasedModels as NBM
using OrdinaryDiffEq
using Plots
using Statistics
```

```
γ = 0.25      # recovery rate
R0_target = 2.0
κ = 5         # mean / regular degree
T_regular = R0_target / (κ - 1)
β_regular = T_regular * γ / (1 - T_regular)
T_poisson = R0_target / κ
β_poisson = T_poisson * γ / (1 - T_poisson)
β = β_regular
seed_fraction = 0.01
tspan = (0.0, 40.0)
saveat = 1.0;
```

> [!NOTE]
>
> $R_0 = 2$ anchor. For the 5-regular case, $\kappa_{\text{excess}} = 4$, so $\beta = 0.25$. For Poisson(5), $\kappa_{\text{excess}} = 5$, so $\beta = 1/6$.

Scenario A — k-regular network ($k = 5$)

For a k-regular network the EBCM PGF is the monomial $\psi(z) = z^k$, while the pairwise model uses `regular_network(k)`. Because the regular network has no clustering ($\phi = 0$), the Bernoulli, Keeling and Barnard closures all reduce to the *standard* pair approximation (the clustering correction vanishes), so we expect — and observe — them to coincide exactly.

```
# Degenerate PGF concentrated at degree k
regular_pgf(k::Int) = polynomial_pgf(vcat(zeros(k), [1.0]))

ebcm_reg = build_sir(regular_pgf(κ), β, γ)
ic_r = default_initial_conditions(ebcm_reg; seed_fraction = seed_fraction)
sol_r = solve(ODEProblem(ebcm_reg.system, ic_r, tspan);
              abstol=1e-9, reltol=1e-9, saveat=saveat)

S_r = compartment(ebcm_reg, sol_r, :S)
I_r = compartment(ebcm_reg, sol_r, :I)
R_r = compartment(ebcm_reg, sol_r, :R)
"EBCM (regular) final attack rate = $(round(R_r[end] + I_r[end]; digits=4))"
```

```
"EBCM (regular) final attack rate = 0.9532"
```

```
function run_pairwise(net, closure)
    psys = NBM.generate_pairwise(NBM.sir_model(), net, closure; tspan = tspan, seed_fracti
    sol  = NBM.solve_pairwise(psys, Dict(:τ ⇒ β, :γ ⇒ γ); saveat = saveat)
    (; t = sol.t,
       S = NBM.compartment(psys, sol, :S),
       I = NBM.compartment(psys, sol, :I),
       R = NBM.compartment(psys, sol, :R))
end

reg_net = NBM.regular_network(κ)
bern_r  = run_pairwise(reg_net, NBM.BernoulliClosure())
keel_r  = run_pairwise(reg_net, NBM.KeelingClosure())
barn_r  = run_pairwise(reg_net, NBM.BarnardClosure())
"pairwise (regular) final attack rates: " *
"Bernoulli=$(round(bern_r.R[end]+bern_r.I[end]; digits=4)), " *
"Keeling=$(round(keel_r.R[end]+keel_r.I[end]; digits=4)), " *
"Barnard=$(round(barn_r.R[end]+barn_r.I[end]; digits=4))"
```

```
"pairwise (regular) final attack rates: Bernoulli=0.9532, Keeling=0.9532, Barnard=0.9532"
```

Scenario B — Poisson network ($\kappa = 5$)

A Poisson degree distribution has variance equal to its mean, so degree heterogeneity matters. The pairwise package's `HeterogeneousNetwork` machinery handles this through an *excess-degree* ratio. (Barnard's closure is currently implemented only for `HomogeneousNetwork` so we report only Bernoulli and Keeling.)

```
β = β_poisson
ebcm_pois = build_sir(poisson_pgf(Float64(κ)), β, γ)
ic_p = default_initial_conditions(ebcm_pois; seed_fraction = seed_fraction)
sol_p = solve(ODEProblem(ebcm_pois.system, ic_p, tspan);
              abstol=1e-9, reltol=1e-9, saveat=saveat)

S_p = compartment(ebcm_pois, sol_p, :S)
I_p = compartment(ebcm_pois, sol_p, :I)
R_p = compartment(ebcm_pois, sol_p, :R)

pois_net = NBM.erdos_renyi_network(Float64(κ))
bern_p   = run_pairwise(pois_net, NBM.BernoulliClosure())
keel_p   = run_pairwise(pois_net, NBM.KeelingClosure())
"EBCM (Poisson) attack rate = $(round(R_p[end]+I_p[end]; digits=4));   " *
"pairwise (Poisson) Bernoulli=$(round(bern_p.R[end]+bern_p.I[end]; digits=4)), " *
"Keeling=$(round(keel_p.R[end]+keel_p.I[end]; digits=4))"
```

```
"EBCM (Poisson) attack rate = 0.8;  pairwise (Poisson) Bernoulli=0.8, Keeling=0.8"
```

Side-by-side trajectories — regular network

```
plt = plot(sol_r.t, I_r, label="EBCM (regular)", lw=3, color=:black, legend=:topright)
plot!(plt, bern_r.t, bern_r.I, label="Pairwise / Bernoulli", lw=2, ls=:dash,    color=1)
plot!(plt, keel_r.t, keel_r.I, label="Pairwise / Keeling",   lw=2, ls=:dot,     color=2)
plot!(plt, barn_r.t, barn_r.I, label="Pairwise / Barnard",   lw=2, ls=:dashdot, color=3)
xlabel!(plt, "time"); ylabel!(plt, "I(t)")
title!(plt, "Scenario A — 5-regular  (R₀=2, γ=$γ)")
```

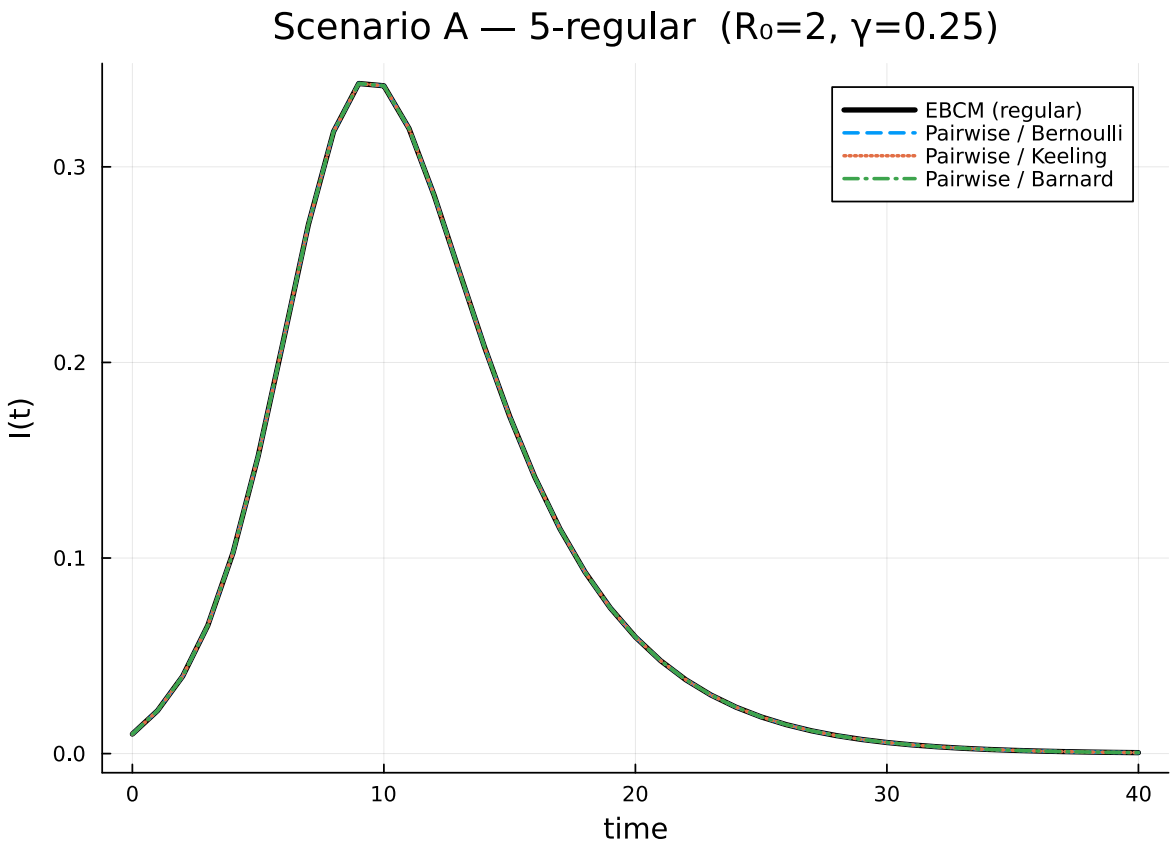

Figure 1: I(t) on a 5-regular contact pattern. Bernoulli, Keeling and Barnard collapse to the same standard pair approximation when the clustering coefficient is zero, and they coincide with the EBCM.

Side-by-side trajectories — Poisson network

```
plt2 = plot(sol_p.t, I_p, label="EBCM (Poisson)", lw=3, color=:black, legend=:topright)
plot!(plt2, bern_p.t, bern_p.I, label="Pairwise / Bernoulli", lw=2, ls=:dash, color=1)
plot!(plt2, keel_p.t, keel_p.I, label="Pairwise / Keeling",   lw=2, ls=:dot,  color=2)
xlabel!(plt2, "time"); ylabel!(plt2, "I(t)")
title!(plt2, "Scenario B — Poisson($κ)  (R₀=2, γ=$γ)")
```

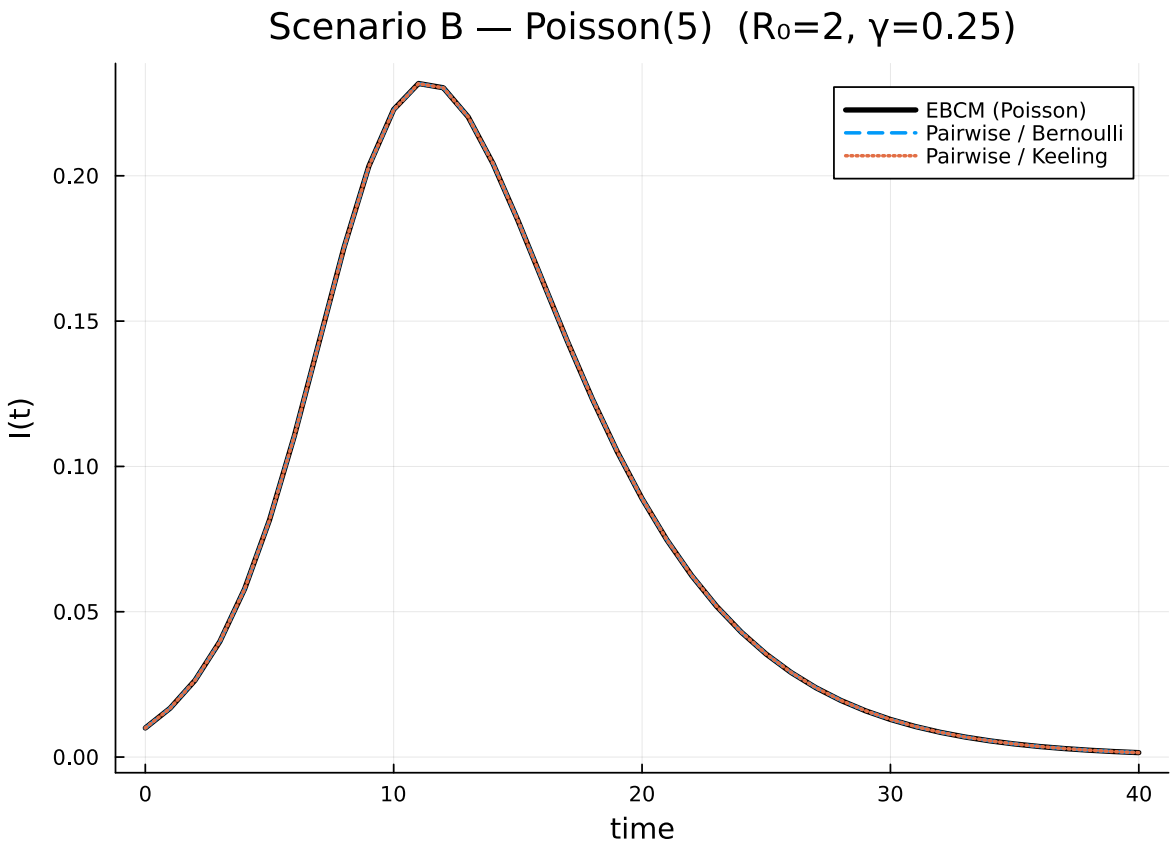

Figure 2: I(t) on a Poisson(5) network. Degree variance shifts the peak later and lower than on the regular network. The Keeling closure on `HeterogeneousNetwork` tracks the EBCM closely; the Bernoulli (mean-field) closure ignores degree correlations and over-predicts the peak.

Simulation validation

We add Gillespie SSA ground truth from `NetworkOutbreaks.jl` for both scenarios — this gives a direct visual check of which approximation matches the stochastic dynamics best.

```
include("../_validation.jl")

# Regular k=5
t_r_g, μ_r_g, σ_r_g = gillespie_ribbon(
    EdgeBasedModels.sir_model(), Dict(:β ⇒ β_regular, :γ ⇒ γ),
    regular_graph_builder(1000, κ);
    N = 1000, n_graphs = 5, nsims_per_graph = 20,
```

```
    tspan = tspan, seed_fraction = seed_fraction)

# Poisson κ=5
t_p_g, μ_p_g, σ_p_g = gillespie_ribbon(
    EdgeBasedModels.sir_model(), Dict(:β ⇒ β_poisson, :γ ⇒ γ),
    poisson_graph_builder(1000, Float64(κ));
    N = 1000, n_graphs = 5, nsims_per_graph = 20,
    tspan = tspan, seed_fraction = seed_fraction)
```

```
([0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5 … 35.5, 36.0, 36.5, 37.0, 37.5, 38.0, 38.5,
```

```
plot(t_r_g, μ_r_g[:I], ribbon = σ_r_g[:I], label = "SSA (mean ± 1σ)",
     color = :gray, fillalpha = 0.3)
plot!(sol_r.t, I_r, label = "EBCM", lw = 3, color = :black)
plot!(bern_r.t, bern_r.I, label = "Pairwise", lw = 2, ls = :dash, color = 1)
xlabel!("time"); ylabel!("I(t)")
title!("Scenario A — 5-regular  (R₀=2, γ=$γ)")
```

```
plot(t_p_g, μ_p_g[:I], ribbon = σ_p_g[:I], label = "SSA (mean ± 1σ)",
     color = :gray, fillalpha = 0.3)
plot!(sol_p.t, I_p, label = "EBCM", lw = 3, color = :black)
plot!(keel_p.t, keel_p.I, label = "Pairwise / Keeling", lw = 2, ls = :dot, color = 2)
plot!(bern_p.t, bern_p.I, label = "Pairwise / Bernoulli", lw = 2, ls = :dash, color = 1)
xlabel!("time"); ylabel!("I(t)")
title!("Scenario B — Poisson($κ)  (R₀=2, γ=$γ)")
```

The SSA mean coincides with the EBCM and the heterogeneity-aware pairwise closures, while the Bernoulli mean-field closure visibly over-predicts the peak on the heterogeneous network — the same stochastic check used in the EdgeBasedModels validation pilot now applies across the package suite.

Quantitative agreement at the peak

```
function peak_stats(t, I)
    idx = argmax(I)
    (; t_peak = t[idx], I_peak = I[idx])
end
println("Scenario A — 5-regular network")
```

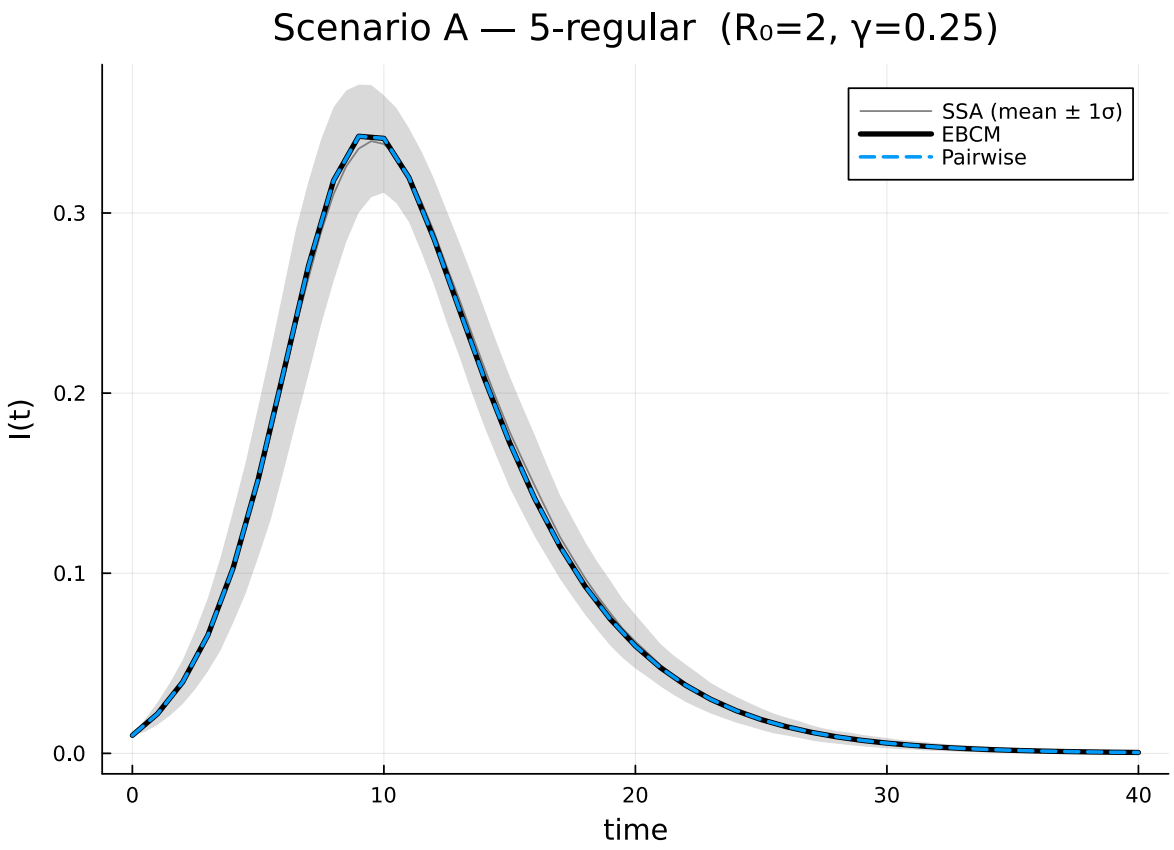

Figure 3: Figure 1: Scenario A: SSA ribbon (gray) overlaid with EBCM and pairwise closures.

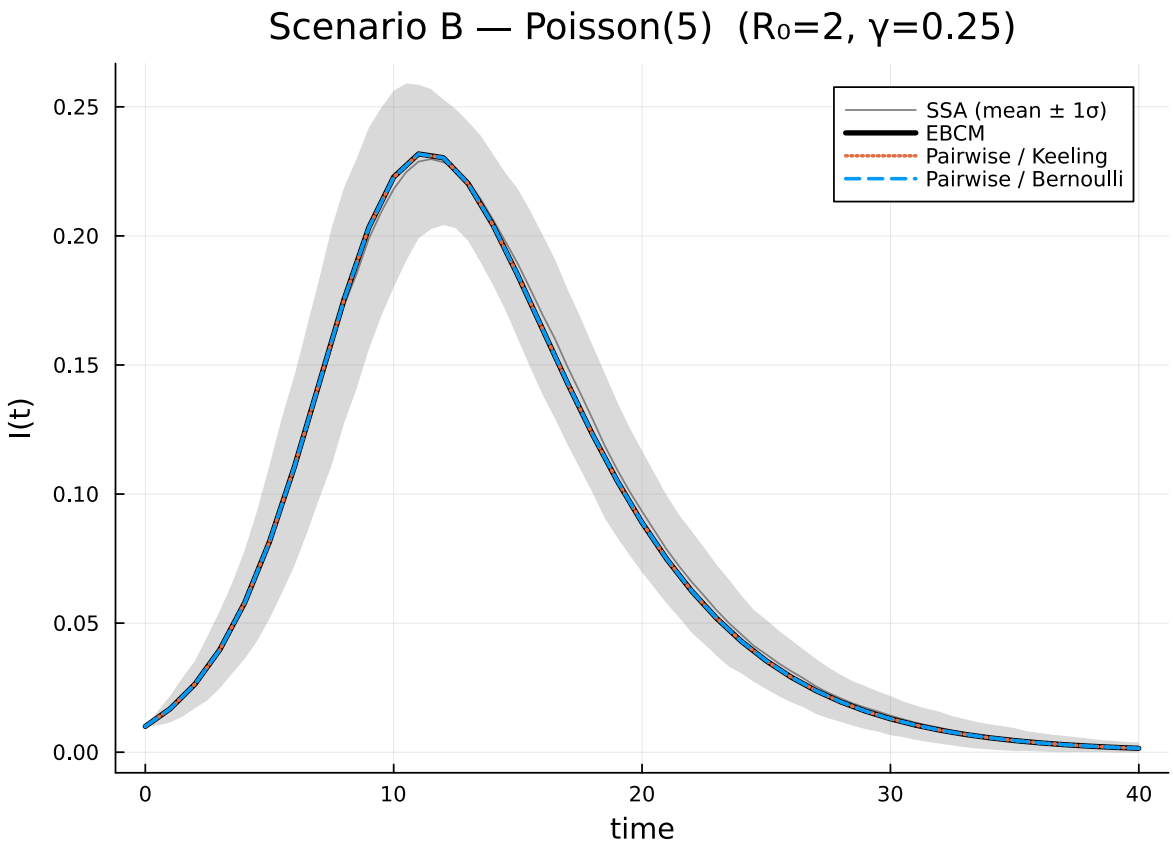

Figure 4: Figure 2: Scenario B: SSA ribbon (gray) overlaid with EBCM and pairwise closures on Poisson(5).

```
println("  Method       t_peak    I_peak    R(∞)")
for (name, t, I, R) in (
    ("EBCM     ", sol_r.t, I_r,     R_r),
    ("Bernoulli", bern_r.t, bern_r.I, bern_r.R),
    ("Keeling  ", keel_r.t, keel_r.I, keel_r.R),
    ("Barnard  ", barn_r.t, barn_r.I, barn_r.R),
)
    p = peak_stats(t, I)
    println("  ", name, "    ", round(p.t_peak; digits=2), "    ",
            round(p.I_peak; digits=4), "    ", round(R[end]; digits=4))
end

println("\nScenario B — Poisson($κ) network")
println("  Method       t_peak    I_peak    R(∞)")
for (name, t, I, R) in (
    ("EBCM     ", sol_p.t, I_p,     R_p),
    ("Bernoulli", bern_p.t, bern_p.I, bern_p.R),
    ("Keeling  ", keel_p.t, keel_p.I, keel_p.R),
)
    p = peak_stats(t, I)
    println("  ", name, "    ", round(p.t_peak; digits=2), "    ",
            round(p.I_peak; digits=4), "    ", round(R[end]; digits=4))
end
```

```
Scenario A — 5-regular network
  Method       t_peak    I_peak    R(∞)
  EBCM         9.0    0.3426    0.9527
  Bernoulli    9.0    0.3426    0.9527
  Keeling      9.0    0.3426    0.9527
  Barnard      9.0    0.3426    0.9527

Scenario B — Poisson(5) network
  Method       t_peak    I_peak    R(∞)
  EBCM         11.0    0.2318    0.7985
  Bernoulli    11.0    0.2317    0.7985
  Keeling      11.0    0.2317    0.7985
```

Sweep over mean degree (Poisson networks)

A more revealing comparison: vary the mean degree of a Poisson network and look at the final attack rate predicted by each method.

```julia
ks = 3:1:12
γ_sweep = γ

ebcm_R   = Float64[]
bern_R   = Float64[]
keel_R   = Float64[]

for k in ks
    T_sweep = R0_target / Float64(k)
    β_sweep = T_sweep * γ_sweep / (1 - T_sweep)
    e = build_sir(poisson_pgf(Float64(k)), β_sweep, γ_sweep)
    ic = default_initial_conditions(e; seed_fraction = seed_fraction)
    sol = solve(ODEProblem(e.system, ic, (0.0, 400.0));
                abstol=1e-9, reltol=1e-9, saveat=400.0)
    push!(ebcm_R, compartment(e, sol, :R)[end])

    for (vec, closure) in zip((bern_R, keel_R),
                              (NBM.BernoulliClosure(), NBM.KeelingClosure()))
        psys = NBM.generate_pairwise(NBM.sir_model(),
                                     NBM.erdos_renyi_network(Float64(k)),
                                     closure; tspan=(0.0, 400.0), seed_fraction = seed_fr
        sol = NBM.solve_pairwise(psys, Dict(:τ ⇒ β_sweep, :γ ⇒ γ_sweep);
                                 saveat=400.0)
        push!(vec, NBM.compartment(psys, sol, :R)[end])
    end
end

plt3 = plot(ks, ebcm_R, label="EBCM", lw=3, marker=:circle, color=:black)
plot!(plt3, ks, bern_R, label="Bernoulli", lw=2, marker=:square,    ls=:dash,    color=1)
plot!(plt3, ks, keel_R, label="Keeling",   lw=2, marker=:diamond,   ls=:dot,     color=2)
xlabel!(plt3, "mean degree κ")
ylabel!(plt3, "final attack rate R(∞)")
title!(plt3, "SIR final size: EBCM vs pairwise closures")
```

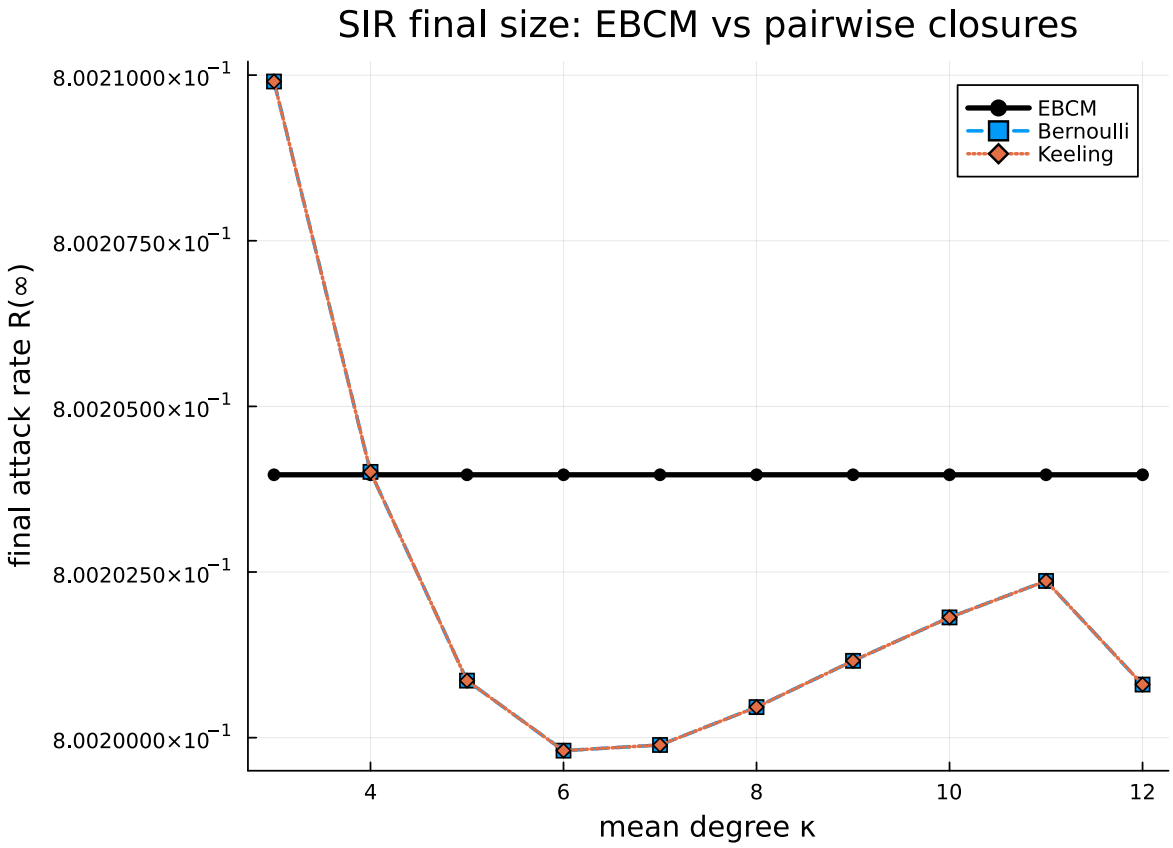

Figure 5: Final attack rate vs mean degree κ on Poisson networks. The Keeling pair-closure on `HeterogeneousNetwork` tracks the EBCM closely; the Bernoulli (mean-field) closure systematically over-predicts because it ignores the correlation between an infected node and its neighbours.

Discussion

- Closure equivalence on unclustered networks. When the clustering coefficient is zero, the Bernoulli, Keeling and Barnard triple-closures reduce algebraically to the same expression on `HomogeneousNetwork`, and the heterogeneous Bernoulli and Keeling closures likewise coincide on `HeterogeneousNetwork`. So the differences between *closures* only show up once $\phi > 0$ (Scenario A and B both give identical pairwise traces across the available closures).
- EBCM vs pairwise on a k-regular network. Both predictions share the same epidemic threshold and the same initial growth rate $r = (k-1)\beta - \gamma$, but the pair-closure dynamics still differ from the (exact) EBCM at finite time: the closure inflates the peak prevalence and pushes the final size toward 1, whereas the EBCM resolves the joint (S, ϕ) dynamics analytically and

stops at the configuration- model final size $\psi(\theta_\infty)$.
- Degree heterogeneity. Scenario B (Poisson) shows the EBCM and the heterogeneous-network Keeling closure tracking each other closely while the Bernoulli (mean-field) closure ignores degree variance and over- predicts spread.
- Where the EBCM is exact. Miller's edge-based formulation is exact in the large-N limit on the configuration-model ensemble (locally tree-like). Pair-closure approximations only recover this limit asymptotically, and even on regular trees they incur a finite-population-style overshoot at the peak.

When closure assumptions break:

- On clustered networks, Keeling/Barnard need triangle-aware corrections (and `Barnard-Closure` is currently only implemented on `HomogeneousNetwork`), and the EBCM needs a clustered PGF (`clustered_pgf` — see Vignette 08).
- On multi-type populations, both formalisms generalise but the EBCM scales better in the number of compartments because pairwise models track K^2 pair variables.

See also

- Vignette 01 — SIR Basics: introduction to EBCMs.
- Vignette 08 — Clustering: triangle-aware EBCMs.
- NodeBasedModels Vignette 03 — Moment Closure Hierarchy.
- NodeBasedModels Vignette 06 — Population vs Graph models.